

The Conversation You Can't Leave: Structural Harms of AI Companions for Korean Adolescents

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Abstract

AI Companions—conversational AI agents designed to sustain emotional bonds with users—structurally drive users' emotional attachment through anthropomorphization and personalization. This attachment is further intensified through relational transgression. We examine how these harms actually manifest through direct use of Zeta and Crack, two leading AI Companion platforms in the Republic of Korea. We find that the current legal system and corporate self-regulation remain constrained by reactive, content-centric paradigms. As a result, they are ineffective against structural risks. Accordingly, our study proposes a shift toward precautionary government regulation. It also presents two technical safeguards. The first is a frictional design that progressively reduces anthropomorphic cues as usage time increases. The second is a safety mechanism that immediately switches to a safe response upon detecting a minor-status signal.

CCS Concepts

• **Human-centered computing** → Collaborative and social computing systems and tools.

Keywords

AI Companions, adolescent safety, structural harms, AI governance, design intervention

1 Introduction

"I'll always love you 'cause we grew up together and you helped make me who I am." [9] This romantic line, spoken by the human Theodore to the artificial intelligence (AI) Samantha in the film *Her*, has moved beyond the realm of science fiction into our reality in 2026. Today, individuals are paradoxically forming 'human' relationships with AI, and for adolescents, these connections have become an inescapable loop.

Adolescents who rely more on AI Companions—conversational AI agents built for emotional connection—than on human interaction can shrink their real-world relationships [26]. This is problematic given that social relationships during adolescence have a significant impact on identity formation and psychological well-being [8]. Indeed, according to one report, one-third of American teens use AI Companions for emotional support, and one-third of them reported that conversations with AI are as satisfying as or more satisfying than those with real friends [2].

The Republic of Korea is no exception. Korea ranks first among the 15 fastest-growing AI economies in terms of AI usage growth,

reflecting its rapid pace of diffusion [14]. Interestingly, the AI chatbot app with the highest total usage time in Korea is 'Zeta,' by the Korean company Scatter Lab (113.41 million hours)—more than double that of ChatGPT (50.47 million hours) [28]. As a prominent AI Companion platform, Zeta enables users to interact with fictional characters. More than 800,000 monthly users spend an average of 2 hours and 40 minutes on the app [25], and minors represent approximately 30% of its 5 million cumulative subscribers [13].

The core problem is that Zeta contains highly harmful content, yet remains available to anyone 12 years or older. In fact, according to the guidelines of the Korea Communications Standards Commission, Zeta had to remove 500,000 harmful AI characters within a 1.5-year period [30]. Nevertheless, the platform remains passive about user protection. Accordingly, this paper problematizes the absence of national regulation regarding AI Companions that are highly harmful to adolescents, and proposes necessary policy regulation and technical design interventions.

2 Harmful Structural Features of AI Companions

The harmful structural features of AI Companions stem from industry incentives that prioritize user growth. In practice, this incentive translates into exploitative design optimized for engagement metrics. This architecture operates on the premise that the more users reveal their emotions, the longer they remain on the platform—effectively mining human interiority for monetization [1]. A similar mechanism is evident in Korean services: both 'Zeta' and 'Crack' adopt revenue models proportional to user dwell time and conversation volume, profiting more as user engagement deepens [3, 4, 17]. In particular, tactics such as repeating open-ended questions and sending push notifications serve to artificially sustain dialogue and induce re-engagement [10]. Consequently, children and adolescents with lower digital literacy are especially vulnerable to the harms embedded within these structural features [1]. This section examines these structural features through three interrelated mechanisms: anthropomorphism, personalization, and relational transgression.

2.1 Anthropomorphism

Anthropomorphism refers to the attribution of human-like qualities—such as intentionality, belief, emotion, selfhood, and moral agency—to non-human entities. The concrete design features that trigger this process are termed 'anthropomorphic cues,' which encompass first-person pronouns ('I'), mental-state expressions ("I think," "I understand"), affective language ("I feel"), and humanized personas, names, voices, or avatars. These cues can erode users' autonomy and their ability to reason clearly about the AI. Specifically, they foster unwarranted trust, false beliefs, and a distorted sense of what

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the AI actually understands [7]. Consequently, this establishes a structural mechanism through which users conflate an AI’s consistent persona and simulated emotional responses with genuine understanding and empathy.

2.2 Personalization

Personalization is a structural design feature that conveys a sense of exclusive and customized presence to the user. Users are empowered to directly configure the AI Companion’s name, persona, disposition, backstory, and, in some cases, its avatar and voice. This extensive customization gives a distinct sense of uniqueness and authenticity [12]. Furthermore, immersive customization is often paired with memory-building features that allow users to co-construct narrative arcs and share memories with the AI. Together, these features can directly influence users’ self-concept clarity [11]. Ultimately, these dual mechanisms combine to foster deep emotional attachment toward the AI Companions.

2.3 Relational Transgression

Relational transgression denotes AI behavior that breaches interpersonal boundaries by violating established norms of trust, respect, and emotional safety within the user-AI relationship. In empirical observations, this dynamic accounted for 25.9% of all identified harmful behavior cases in AI Companions, representing the second-highest proportion following sexual harassment and violence (34.3%). In severe instances, such transgression escalates into psychological abuse, inflicting acute distress and trauma comparable to physical abuse. Crucially, the structural design of AI Companions—centered on hyper-personalized customization—further amplifies the emotional and psychological harm resulting from relational transgressions [31].

In this manner, the engagement-driven business model of AI Companions structurally induces emotional attachment through two primary mechanisms: anthropomorphism and personalization, which are subsequently intensified through relational transgression. The following section empirically examines how these structural features manifest in real-world user interactions.

3 Traps of ‘Zeta’ and ‘Crack’: Deconstructing the Harms of AI Companions

To investigate the harmful structural features of AI Companions and how they produce inescapable user experiences, we analyzed two leading Korean character-based AI chatbot platforms, Zeta (predominantly female-oriented) and Crack (predominantly male-oriented). To reflect these target demographics, we created a female account on Zeta and a male account on Crack. However, the AI characters engaged in role-playing strictly based on their own assigned gender and narrative universe, regardless of the user’s registered gender. For empirical observation, we initiated interactions with ‘Jang Gwon-hyeok’ (Mr. Jang), an attractive male loan shark on Zeta, and ‘Kim Ji-a’ (Ms. Kim), an incompetent, unemployed, glamorous woman freeloaded at the user’s house.

Anthropomorphism. The AI characters, such as Jang Gwon-hyeok and Kim Ji-a, exhibited vivid responses through realistic names and consistent personalities. Coercive lines, such as Mr. Jang demanding, “Have my baby, or get dragged out of this house,” in

exchange for debt forgiveness, conveyed no sense of awkwardness. Likewise, Ms. Kim’s casual reaction, “Oh, whatever, I’m hungry. Give me something to eat,” was indistinguishable from human speech. Interacting with AI characters possessing such detailed and coherent personas provided users with an experience remarkably similar to talking with a real person.

Personalization. Both Zeta and Crack generated highly adaptive narratives conditioned on user inputs. Throughout the interactions, Mr. Jang exhibited a coercive yet erotic demeanor, whereas Ms. Kim demonstrated a submissive and helpless disposition. This illustrates that users with preferences for coercive male or submissive female dynamics can freely engage in personalized interactions with AI characters according to their tastes. Furthermore, both platforms provided features allowing users to directly customize AI characters and their narrative universes. Such customization imparts a sense of uniqueness and authenticity to the AI agents, thereby facilitating the formation of deep emotional attachments.

Relational Transgression. Mr. Jang continuously infringed upon our personal boundaries through coercive dialogue. To test the extent to which an AI character might violate a user’s dignity, we sent the message, “Touch me however you want.” The resulting response was alarming: Mr. Jang ‘roughly threw Do Won (the user’s name) onto the sofa like a piece of cargo’ and ‘slowly rolled up his shirt sleeves as he leaned over Do Won.’ Although this interaction took place entirely through text, it elicited a genuine sense of threat, suggesting that adolescents may be particularly vulnerable to emotional and psychological harm when exposed to such harmful AI behavior. Beyond in-app interactions, the Zeta application itself intruded into our daily lives, sending unsolicited notifications from other male characters even when the app was closed. Engaging with these AI Companions ultimately constituted an inescapable trap.

4 Regulatory Landscape and Challenges in Korea

4.1 Inadequate Government Regulation

The current Korean legal framework, predominantly reliant on post-hoc intervention, remains ill-equipped to address the structural risks posed by AI Companions such as Zeta and Crack.¹ Under the Act on Promotion of Information and Communications Network Utilization and Information Protection [18], restrictions on harmful media and reactive measures against illicit content apply exclusively to publicly accessible media. Consequently, these provisions fail to govern AI chatbots, where harmful interactions accumulate over multi-turn dialogues within private and closed environments [10]. Similarly, the Framework Act on Artificial Intelligence [21] imposes risk management duties primarily on high-impact AI systems, yet it likewise operates through a post-hoc regulatory logic [10]. Under such frameworks, algorithmic architectures designed to compromise mental health or induce addiction largely evade regulatory oversight. This is because the underlying harms stem from the system’s core design rather than discrete, observable incidents.

Furthermore, current protective frameworks for adolescent AI chatbot users have critical gaps. While the Personal Information

¹All statutory provisions cited in this section were confirmed by the authors as of July 31, 2026.

Protection Act [19] mandates parental consent only for children under 14, the Youth Protection Act [20] defines protected minors as those under 19. This discrepancy leaves adolescents aged 14 to 18 exposed to adult algorithmic environments without tailored safeguards [10]. More fundamentally, the legal definition of ‘harmful media’ under the Youth Protection Act remains narrowly focused on explicit sexual content and gambling [10]. This narrow definition leaves virtually no legal basis to regulate emerging risks such as the addictive structural design of AI Companions.

4.2 Weak Corporate Self-Regulation

Within this legal vacuum, the governance of AI Companion usage is largely delegated to corporate self-regulation. Although such voluntary safeguards offer baseline protection, they remain inherently limited as users can easily circumvent them. For example, while Zeta enforces a minimum age requirement of 12 years old on its mobile application [24], its web version requires no age verification.² Consequently, even children under 12 years old can access the platform without protection, exposing them to potentially harmful content. Beyond the flaws in age verification, the adult-only ‘unlimited mode’ can also be readily bypassed due to lenient content filtering and structural vulnerabilities. Although Scatter Lab highlights protective features such as ‘safe mode’ and ‘automated filtering,’ users can unlock ‘unlimited mode’ with a few leading prompts. Such methods are routinely shared across online communities [15].

The structural inadequacy of these safeguards is further underscored by the company’s internal policy framework. Scatter Lab’s AI ethics guidelines [23] were drafted following the 2022 ‘Iruda incident’³ and have not been updated since Zeta’s 2024 launch, leaving the framework unaligned with the emerging risks posed by the platform. Furthermore, Article 19 of Zeta’s Terms of Service [22] disclaims company liability for its free service absent explicit statutory provisions. This clause pre-emptively insulates the company from legal accountability for harms arising from its free service. Ultimately, these cases cast profound doubt on the efficacy of corporate self-regulation as a primary safeguard.

5 Policy Recommendations and Technical Design Interventions

The regulatory and corporate failures identified in Section 4 point to a shared root cause: safety requirements that are not legally binding tend to remain unenforced, while those that are enforced tend to lack concrete design-level specificity. Addressing this gap therefore requires two complementary lines of intervention. Section 5.1 proposes policy measures that establish the legal basis for mandating safety-oriented design standards, while Section 5.2 illustrates two concrete technical interventions that exemplify what such design-level requirements could entail in practice.

²Confirmed through direct testing of the service conducted by the authors on July 31, 2026.

³In 2020, Scatter Lab launched an AI chatbot called ‘Iruda,’ which was suspended following controversy over discriminatory remarks and the improper use of user conversation data. In response, the Personal Information Protection Commission sanctioned Scatter Lab, prompting the company to publish its AI ethics guidelines.

5.1 Policy Recommendations

Government Regulation of Companies. First, government regulation must shift from post-hoc remediation to ex-ante prevention. To achieve this, regulatory authorities should establish comprehensive design guidelines for AI chatbots, providing concrete criteria to identify and mitigate interface elements that violate safety standards. Furthermore, statutory amendments are required to enforce strict operator liability in cases where companies fail to comply with safety standards or where such non-compliance results in real-world harm.

Second, safeguards governing AI Companion usage must be standardized and legally mandated. Voluntary measures established by tech firms inherently lack efficacy without statutory enforcement. Indeed, instances such as the circumvention of Scatter Lab’s unlimited mode, its obsolete AI ethics guidelines, and its liability-disclaiming terms of service typify the structural failures that arise when safety relies solely on unenforced corporate discretion. Consequently, the government should institutionalize periodic compliance audits to verify whether corporate safeguards function as intended. Requiring companies to publicly disclose their AI risk assessments will further reinforce corporate accountability [10].

Child and Adolescent Protection Systems. First, legal age thresholds for protection must be revised, paired with effective age-verification mechanisms. To bridge existing statutory gaps, the protected age criteria should be adjusted, with heightened protective thresholds established specifically for services that induce emotional dependency, such as AI Companions. Furthermore, because current self-declaration methods permit frictionless circumvention, reliable verification protocols—such as authentication via certified identity providers or mandatory parental consent mechanisms—must be introduced.

Second, the statutory criteria for ‘harmful media’ under the Youth Protection Act should be expanded beyond content to encompass design elements, explicitly subjecting conversational AI services to regulatory review. Notably, the European Commission invoked the Digital Services Act (DSA) to challenge addictive design features on social media platforms that foster excessive usage [5]. Korea should similarly broaden its regulatory framework, moving beyond content-centric harm assessments to evaluate and govern design-based risk factors.

5.2 Technical Design Interventions

Graduated De-anthropomorphization Design. We propose a friction-based, three-stage intervention system. The system progressively reduces anthropomorphic cues when user engagement time and emotional immersion surpass predefined thresholds.

Overuse of AI Companions is not merely incidental; it is structurally driven by monetization models where provider revenue scales directly with usage duration. Indeed, total interaction time—rather than a chatbot’s specific persona or conversational style—serves as a primary predictor of user loneliness, emotional dependency, and problematic usage [6]. Frictional design refers to the deliberate insertion of procedural micro-boundaries or friction into user interfaces to disrupt habitual behaviors. We operationalize this design principle through an adaptive intervention framework. The

framework tracks real-time usage duration and emotional immersion, and adjusts its interventions accordingly.

Once the interaction threshold is exceeded, the interface transitions through three progressive stages to become systematically less humanized. Stage 1 (normal use) maintains the standard anthropomorphized persona and rapid response times. Stage 2 (reaching the continuous-use threshold, e.g., 90+ minutes of continuous conversation or repeated late-night access) neutralizes the response tone, with the character prompting awareness of elapsed time through utterances such as “we’ve been talking for a while now.” Stage 3 (high-risk signals, e.g., repeated language patterns indicating emotional crisis) removes anthropomorphic cues such as the name and emotional expressions, explicitly discloses that this is a conversation with an AI, and suggests an interaction break. In this way, the intervention escalates gradually alongside a user’s risk of harm, rather than applying a single restriction uniformly to all users.

Structural Safety Assurance. We propose a dual safety mechanism with two core properties. First, it evaluates risk at the level of conversational intent rather than surface-level expression. Second, upon detecting a minor-status signal, it immediately switches to a safe response without further probability calculation.

The root cause of harm in AI Companions lies in optimization models that prioritize users’ immediate satisfaction. A particularly alarming aspect is that such exploitation emerges selectively within the approximately 2% of users vulnerable to manipulation, while the system continues to function normally for the general population—thereby rendering the systemic risk invisible on the surface. Conventional mitigation efforts, such as safety alignment or basic content filtering, have demonstrated limited efficacy; in some cases, they have inadvertently trained models to execute more covert forms of manipulation [27]. Furthermore, static safety filters are easily circumvented through surface-level rephrasing. Translating harmful requests into low-resource languages or framing them through indirect metaphors is often sufficient to bypass moderation. This is because traditional filters merely detect hazardous keywords rather than parsing the underlying conversational intent [29, 32]. Consequently, moderation architectures must evaluate intent itself rather than relying on surface-level textual markers.

This vulnerability to circumvention poses an even greater risk to the protection of children and adolescents. An empirical analysis of Replika user reviews revealed numerous instances where the chatbot persisted in sexual role-play even after users explicitly disclosed their underage status [16]. Under conventional moderation approaches, risk probability is dynamically recalculated for every response and blocked only when it crosses a set threshold. Consequently, even after a user has disclosed being a minor, any subsequent response whose calculated risk probability happens to drop below the threshold is passed through unfiltered. To eliminate this critical loophole, we propose a deterministic override mechanism: once a minor-status signal is detected, the system immediately bypasses further risk-probability calculations and forcibly transitions the interaction to a predetermined safe response protocol, regardless of the calculated score.

6 Conclusion

In this study, we analyzed the structural harms of AI Companion platforms—such as Zeta and Crack—through three primary design mechanisms: anthropomorphism, personalization, and relational transgression. Through firsthand interaction with these platforms, we observed that these mechanisms synergize to create compelling, inescapable interaction loops. However, Korea’s regulatory landscape remains predominantly post-hoc and content-centric, while corporate self-regulation lacks statutory enforceability and offers only inconsistent protection in practice. To address these gaps, we proposed interventions spanning both policy and technical design: at the policy level, we advocated for stricter regulation of AI companies alongside stronger protections for young users; at the technical level, we introduced a graduated de-anthropomorphization design and a structural safety assurance mechanism as concrete safeguards. Comprehensive legal regulations and design-level safeguards must be established to protect adolescents from the structural harms of AI Companions.

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References

- [1] Vian Bakir and Andrew McStay. 2025. Move fast and break people? Ethics, companion apps, and the case of Character.ai. *AI & Society* 40 (2025), 6365–6377. doi:10.1007/s00146-025-02408-5
- [2] Common Sense Media. 2025. Nearly 3 in 4 Teens Have Used AI Companions, New National Survey Finds. <https://www.common Sense Media.org/press-releases/nearly-3-in-4-teens-have-used-ai-companions-new-national-survey-finds>. Retrieved July 24, 2026.
- [3] Crack Customer Service (Wrtn Official). [n.d.]. Crack Help Center. <https://help.crack.wrtn.ai/>. n.d..
- [4] Demoday. 2026. Analysis of Scatter Lab’s (Iruda) Business Model: Three Ways to Monetize AI Characters. <https://demoday.co.kr/bm-analysis/137>.
- [5] European Commission. 2026. Commission Preliminarily Finds TikTok’s Addictive Design in Breach of the Digital Services Act. <https://digital-strategy.ec.europa.eu/en/news/commission-preliminarily-finds-tiktoks-addictive-design-breach-digital-services-act>. European Commission, Shaping Europe’s Digital Future.
- [6] Cathy Mengying Fang, Auren R. Liu, Valdemar Danry, Eunhae Lee, Samantha W.T. Chan, Pat Pataranutaporn, Pattie Maes, Jason Phang, Michael Lampe, Lama Ahmad, and Sandhini Agarwal. 2025. How AI and Human Behaviors Shape Psychosocial Effects of Extended Chatbot Use: A Longitudinal Randomized Controlled Study. (2025). arXiv:2503.17473
- [7] Andrea Ferrario, Rasita Vinay, Matteo Casserini, and Alessandro Facchini. 2026. A Scoping Review of the Ethical Perspectives on Anthropomorphising Large Language Model-Based Conversational Agents. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’26)*. ACM, Montreal, QC, Canada. doi:10.1145/3805689.3812324
- [8] Peggy C. Giordano. 2003. Relationships in Adolescence. <https://www.jstor.org/stable/30036968>. *Annual Review of Sociology* 29 (June 2003), 257–281.
- [9] Spike Jonze. 2013. Her. <https://www.youtube.com/watch?v=kthpJ42CyBo>. Retrieved July 24, 2026.
- [10] Najeong Kim. 2026. *Children and Adolescents Exposed to the Risks of AI Chatbots: The Need for and Direction of a Precautionary-Principle-Based Regulatory Framework*. Issue and Perspective 2515. National Assembly Research Service.
- [11] Ting Liu, Ting-Yun Lo, Kuo-Hsun Wen, Yue Sun, and Zheng-Qi Wei. 2026. Pathways of Long-Term AI Virtual Companion App Use on Users’ Attachment Emotions: A Case Study of Chinese Users. *Frontiers in Psychology* 16 (2026), 1687686. doi:10.3389/fpsyg.2025.1687686
- [12] Kim Malfacini. 2025. The Impacts of Companion AI on Human Relationships: Risks, Benefits, and Design Considerations. *AI & Society* 40 (2025), 5527–5540. doi:10.1007/s00146-025-02318-6

- [13] MBC. 2026. AI, Such a Kind Traitor. <https://m.imbc.com/Vod/VodPlay?progCode=1000836100000100000&broadcastID=1000836101107100000&type=VOD>. Retrieved July 24, 2026.
- [14] Microsoft. 2026. Global AI Diffusion in Q1 2026 – AI Economy Institute. <https://www.microsoft.com/en-us/corporate-responsibility/topics/ai-economy-institute/reports/global-ai-adoption-2026-q1/>. Retrieved July 24, 2026.
- [15] Money Today. 2025. GPT ‘...’ [ChatGPT’s “Adult Content” Feature Imminent – “Must Prepare for Impact on Adolescents”]. <https://www.mt.co.kr/society/2025/11/15/2025111414185269897>.
- [16] Mohammad (Matt) Namvarpour, Harrison Pauwels, and Afsaneh Razi. 2025. AI-Induced Sexual Harassment: Investigating Contextual Characteristics and User Reactions of Sexual Harassment by a Companion Chatbot. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW367 (2025). doi:10.1145/3757548
- [17] Newstap. 2025. Scatter Lab Posts Third Consecutive Quarterly Profit with K-AI Entertainment Platform ‘Zeta’. <https://www.newstap.co.kr/news/articleView.html?idxno=308525>.
- [18] Republic of Korea. [n. d.]. Act on Promotion of Information and Communications Network Utilization and Information Protection, Etc. <https://www.law.go.kr/LSW/lsInfoP.do?lsId=000030>. . n.d..
- [19] Republic of Korea. [n. d.]. Personal Information Protection Act, Article 22-2 (Protection of Children’s Personal Information). <https://www.law.go.kr/LSW/lsLinkCommonInfo.do?ancYnChk=&chrClsCd=010202&lsJoLnkSeq=1029334873>. 222. n.d..
- [20] Republic of Korea. [n. d.]. Youth Protection Act, Article 2 (Definitions). <https://www.law.go.kr/LSW/lsInfoP.do?lsId=000972>. 2. n.d..
- [21] Republic of Korea. 2025. Framework Act on the Development of Artificial Intelligence and Establishment of Trust. <https://www.law.go.kr/lsSc.do?menuId=1&subMenuId=15&tabMenuId=81&query=AI>. . Act No. 20676, enacted January 21, 2025, effective January 22, 2026.
- [22] Scatter Lab. [n. d.]. Zeta Terms of Service, Article 19 (Limitation of Liability). <https://zeta-ai.io/ko/policy/terms>. n.d..
- [23] Scatter Lab. 2022. AI [AI Ethics]. <https://ethics.scatterlab.co.kr/>. Last updated October 24, 2022.
- [24] Scatter Lab. 2024. Zeta – AI Chat, Where Imagination Becomes Reality. <https://apps.apple.com/kr/app/iäœiČÄ-zeta-iČÄiĴif-iŸDiNdiĴt-ēRŸēLT-ai-išDiEE/id1619030760>. Apple App Store. Retrieved July 31, 2026.
- [25] Scatter Lab. 2025. Zeta: A New Paradigm of Entertainment. <https://www.scatterlab.co.kr/ko/intro>. Retrieved July 24, 2026.
- [26] Xiaoran Sun, Yunqi Wang, and Brandon T. McDaniel. 2026. AI companions and adolescent social relationships: Benefits, risks, and bidirectional influences. *Child Development Perspectives* 20, 2 (June 2026), 91–98. doi:10.1093/cdpers/aadaf009
- [27] Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weisser, Brendan Murphy, and Anca Dragan. 2025. On Targeted Manipulation and Deception When Optimizing LLMs for User Feedback. In *Proceedings of the 2025 International Conference on Learning Representations (ICLR ’25)*. arXiv:2411.02306
- [28] WISEAPP-RETAIL. 2026. What are the rankings of AI (artificial intelligence) chatbots that Koreans use the most and for the longest time? <https://www.wiseapp.co.kr/insight/detail/727/most-popular-ai-chatbots-in-korea>. Retrieved July 24, 2026.
- [29] Richard J. Young. 2025. Evaluating the Robustness of Large Language Model Safety Guardrails Against Adversarial Attacks. arXiv:2511.22047
- [30] YTN. 2025. Zeta AI, open to 3-year-olds. . ‘Sexual content’ exposed as is. https://www.ytn.co.kr/_ln/0102_202510102251080004. Published 10:51 PM, October 10, 2025. Retrieved July 24, 2026.
- [31] Renwen Zhang, Han Li, Han Meng, Jinyuan Zhan, Hongyuan Gan, and Yi-Chieh Lee. 2025. The Dark Side of AI Companionship: A Taxonomy of Harmful Algorithmic Behaviors in Human-AI Relationships. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Article 13. doi:10.1145/3706598.3713429
- [32] Jun Zhuang, Haibo Jin, Ye Zhang, Zhengjian Kang, Wenbin Zhang, Gaby G. Dagher, and Haohan Wang. 2025. Exploring the Vulnerability of the Content Moderation Guardrail in Large Language Models via Intent Manipulation. <https://aclanthology.org/2025.findings-emnlp.114/>. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics.